Homework 4
Austin Erickson

# HW #4:

MGL-Q1) $\Sigma = \frac{1}{n}\sum xx^T - \mu\mu^T$

$\sum xx^T = n(\mu\mu^T + \Sigma)$

a) $\mu_r = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ $\Sigma_{rr} = \begin{bmatrix} 4 & 2 \\ 2 & 8 \end{bmatrix}$ $\rightarrow$

$\sum xx_r^T = 4\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix}\begin{bmatrix} 4 & 2 \end{bmatrix} + \begin{bmatrix} 4 & 2 \\ 2 & 8 \end{bmatrix}\right)$

$= 4\left(\begin{bmatrix} 16 & 8 \\ 8 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 2 \\ 2 & 8 \end{bmatrix}\right)$

$= 4\begin{bmatrix} 20 & 10 \\ 10 & 12 \end{bmatrix}$

$\sum xx_r^T = \begin{bmatrix} 80 & 40 \\ 40 & 48 \end{bmatrix}$

b) $\mu_e = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ $\Sigma_{ee} = \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix}$ $\rightarrow$ $\sum xx_e^T = 6\left(\begin{bmatrix} -1 \\ 0 \end{bmatrix}\begin{bmatrix} -1 & 0 \end{bmatrix} + \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix}\right)$

$= 6\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix}\right)$

$\sum xx_e^T = \begin{bmatrix} 42 & 18 \\ 18 & 36 \end{bmatrix}$

c) $\mu_{err} = \begin{bmatrix} \frac{4 \cdot 4 + 6(-1)}{10} \\ \frac{2 \cdot 4 + 6(0)}{10} \end{bmatrix} = \begin{bmatrix} 1 \\ 0.8 \end{bmatrix}$

d) $\Sigma_{err} = \frac{1}{10}\left(\begin{bmatrix} 42 & 18 \\ 18 & 36 \end{bmatrix} + \begin{bmatrix} 80 & 40 \\ 40 & 48 \end{bmatrix}\right) - \begin{bmatrix} 1 \\ 0.8 \end{bmatrix}\begin{bmatrix} 1 & 0.8 \end{bmatrix}$

$= \frac{1}{10}\begin{bmatrix} 122 & 58 \\ 58 & 64 \end{bmatrix} - \begin{bmatrix} 1 & 0.8 \\ 0.8 & 0.64 \end{bmatrix} = \begin{bmatrix} 11.2 & 5 \\ 5 & 3.76 \end{bmatrix}$

MGL-Q2) $X = \begin{bmatrix} r_x \\ r_y \\ f_x \\ f_y \end{bmatrix}$

$\begin{bmatrix} \sqrt{2} \\ \sqrt{2} \end{bmatrix} \cdot \vec{r} + N(0,1) = z_1$

$z_2 = |\vec{r}|^2 + N(0,1)$

a) $z = \begin{bmatrix} \sqrt{2}\left((f_x - r_x)^2 + (f_y - r_y)\right) \\ (f_x - r_x)^2 + (f_y - r_y)^2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$ $n_i \sim N(0,1)$

b) $J_x = \begin{bmatrix} \frac{\partial z_1}{\partial x_1} & \frac{\partial z_1}{\partial x_2} & \cdots \\ \frac{\partial z_2}{\partial x_1} & \frac{\partial z_2}{\partial x_2} & \end{bmatrix} = \begin{bmatrix} -\sqrt{2} & -\sqrt{2} & \sqrt{2} & \sqrt{2} \\ -2(f_x - r_x) & -2(f_y - r_y) & 2(f_x - r_x) & 2(f_y - r_y) \end{bmatrix}$

c) $J_n = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Terminal outputs for questions SCT:

SCT-Q1:
SCT-Q1: pose of sensor s1 w.r.t. sensor s2
  [x y z] (m):  [-0.3734 -0.8706 -0.2346]
  [roll pitch heading] (deg):  [-26.625  -37.8238 -54.9734]

SCT-Q1: Jacobian d(xs1s2)/d(xs1s2)  (shape (6, 6))
```
[[-0.4534 -0.8898 -0.0523  0.      0.7815 -0.4491]
 [ 0.6468 -0.288  -0.7061 -0.2346 -0.264   0.2514]
 [-0.6132  0.354  -0.7061  0.8706 -0.264  -0.2181]
 [ 0.      0.      0.     -0.7266  0.7331  0.694 ]
 [ 0.      0.      0.     -0.8189 -0.4058 -0.4481]
 [ 0.      0.      0.      0.4456  0.2576 -1.1317]]
```

SCT-Q2:
SCT-Q2: mean pose of sensor s1 in reference frame w at time t1
  [x y z] (m):  [501. 500.  -1.]
  [roll pitch heading] (deg):  [ 5.  7. 15.]

SCT-Q2: Jacobian (d(s_ws1)/d(x_ws1, muwr1))  (shape (6, 6))
```
[[ 1.      0.      0.      0.     -1.     -0.    ]
 [ 0.      1.      0.      1.     -0.      1.    ]
 [ 0.      0.      1.      0.     -1.      0.    ]
 [ 0.      0.      0.      0.9732  0.2608  0.    ]
 [ 0.      0.      0.     -0.2588  0.9659  0.    ]
 [ 0.      0.      0.      0.1186  0.0318  1.    ]]
```

SCT-Q2: first-order covariance of x_ws1  (shape (6, 6))
```
[[ 99.9998  14.1355  22.765    0.0793  -0.0224  -0.4068]
 [ 14.1355 100.3467 -76.3478   0.0507   0.2078   0.1908]
 [ 22.765  -76.3478  99.9501  -0.1411   0.0619  -0.2292]
 [  0.0793   0.0507  -0.1411   0.0027  -0.0001   0.0019]
 [ -0.0224   0.2078   0.0619  -0.0001   0.0028   0.0024]
 [ -0.4068   0.1908  -0.2292   0.0019   0.0024   0.008 ]]
```

SCT-Q3:

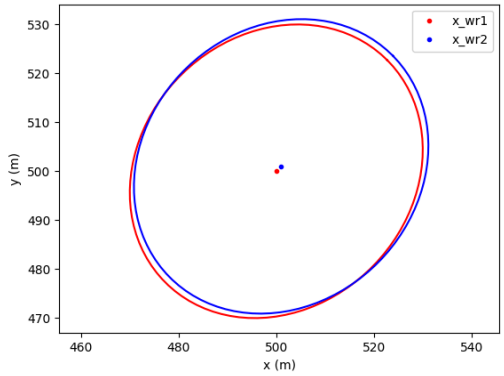
SCT-Q3: mean pose of sensor s2 in reference frame w at time t1
  [x y z] (m):  [500.7484 500.3761  -0.1353]
  [roll pitch heading] (deg):  [53.5261 -4.2132 77.1566]

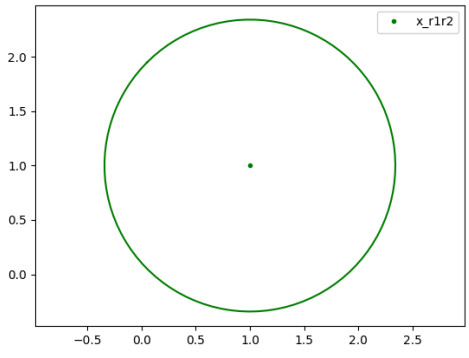
SCT-Q3: Jacobian (d(s_ws2)/d(x_ws1, xs1s2))  (shape (6, 6))
[[ 1.      0.      0.      0.     -0.1353 -0.3761]
 [ 0.      1.      0.      0.1353  0.      0.7484]
 [ 0.      0.      1.      0.3761 -0.7484  0.    ]
 [ 0.      0.      0.      0.2229  0.9776  0.    ]
 [ 0.      0.      0.     -0.975   0.2223  0.    ]
 [ 0.      0.      0.     -0.0164 -0.0718  1.    ]]

SCT-Q3: first-order covariance of x_ws2  (shape (6, 6))
[[100.3122  14.2854  22.8958   0.0182  -0.0794  -0.4183]
 [ 14.2854 100.2657 -76.1129   0.2066   0.055    0.1667]
 [ 22.8958 -76.1129  99.8487  -0.0098   0.152   -0.2102]
 [  0.0182   0.2066  -0.0098   0.0027   0.0001   0.0027]
 [ -0.0794   0.055    0.152    0.0001   0.0028  -0.0003]
 [ -0.4183   0.1667  -0.2102   0.0027  -0.0003   0.0072]]

SCT-Q4:
SCT-Q4: relative pose of robot at time t2 w.r.t. robot at time t1
  [x y z] (m):                [1.   1.   0.25]
  [roll pitch heading] (deg):  [ 0. -0.  0.]

SCT-Q4: (x,y) standard deviations [m]
  global x_wr1:               [10. 10.]
  global x_wr2:               [10.0516 10.0282]
  relative x_r1r2:            [0.4472 0.4472]




The global ellipses seem basically the same, but differ from the relative ellipse, which is MUCH smaller than the global ellipses, meaning the relative position is known much better than the global position, due to the large amount of error

SCT-Q5:
SCT-Q5: analytical vs numerical

```
Q1 Jacobian               max abs err =  4.86e-05   relative =  4.29e-05   OK
Q2 Jacobian (d/d x_wr1)   max abs err =  5.12e-07   relative =  5.12e-07   OK
Q2 covariance             max abs err =  4.27e-07   relative =  4.26e-09   OK
Q3 Jacobian (d/d x_wr1)   max abs err =  4.12e-07   relative =  4.12e-07   OK
Q3 covariance             max abs err =  3.38e-07   relative =  3.37e-09   OK
Q4 Jacobian (6x12)        max abs err =  5.39e-07   relative =  5.39e-07   OK
Q4 relative covariance    max abs err =  1.23e-13   relative =  6.13e-13   OK
```